Software Requirements Specification

E-Commerce Web Application

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This working draft uses the introduction, overall description, specific requirements and supporting appendices approach described by the installed `srs-documentation` skill. It groups specific requirements into Sections 3–8 for readability. Requirements are proposed for review; their inclusion does not mean they have passed acceptance testing. Replace bracketed placeholders before submission. No formal standards certification is claimed.

Revision history

Version	Date	Description	Author
0.1	4 November 2026	Initial outline, introduction and requirement examples	[Student name]
0.2	7 November 2026	Applied SRS skill guidance; expanded requirements, interfaces and review workflow using current project sources	[Student name]

Version	Date	Description	Author
0.3	7 November 2026	Expanded use cases, data rules, quality criteria, state model and complete draft requirement traceability	[Student name]

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1. Introduction

1.1 Purpose

This Software Requirements Specification defines the expected behaviour and quality requirements of the E-Commerce Web Application. It describes the facilities provided to guests, registered customers and administrators, together with the system's external interfaces, data requirements and constraints. The document provides a common reference for project evaluation, development and testing.

1.2 Project Scope

The E-Commerce Web Application provides a browser-based platform through which customers can discover products, manage a shopping cart and wishlist, maintain delivery addresses, place orders and view order progress. Customers can choose online payment through Razorpay or Cash on Delivery. Administrators manage the product catalog, inventory, orders and user roles through a restricted administration area.

The draft scope includes:

- Account registration, sign-in, profile management and password changes.
- Product browsing, search, category and price filters, sorting and product details.
- Product images, selling prices, original prices and applicable discounts.
- Shopping cart quantities and selection of items for checkout.
- Saved delivery addresses and a separate delivery PIN location lookup.
- Online payment, Cash on Delivery, order history and eligible order cancellation.
- Wishlists, product reviews and rating summaries.
- Administrative catalog, stock, order, user and system-health functions.

Proposed exclusions to confirm with the project guide: a multi-vendor seller portal, live courier GPS tracking, customer wallets and native Android/iOS applications. A responsive website is included.

1.3 Intended Audience

The intended readers are the student project team, project guide, academic evaluators, developers and testers.

1.3.1 Objectives and Document Overview

The project aims to let customers purchase selected catalog items, preserve accurate order and stock records, and let administrators maintain the catalog and fulfil orders. Sections 1–2 explain the scope and environment. Sections 3–6 specify behaviour, interfaces, quality and data requirements. Sections 7–8 connect user workflows to acceptance checks. Section 9 records decisions required before submission.

1.4 Definitions and Abbreviations

Term	Meaning
SRS	Software Requirements Specification
FR	Functional Requirement: behaviour the system must provide
NFR	Non-Functional Requirement: a quality or constraint the system must meet
COD	Cash on Delivery

Term	Meaning
INR	Indian Rupee
API	Application Programming Interface
PIN	Postal Index Number used for Indian postal locations
Acceptance criterion	An observable condition used to decide whether a requirement is satisfied
TBD	To be determined before the document is finalised

1.5 Project References

- Project memory and current decisions.
- Frontend plan.
- Original backend plan.
- Current application source under `Frontend/src`, `Backend/src` and `Backend/prisma/schema.prisma`.

Historical plans may describe older behaviour. Check the current project decisions and relevant source before turning a feature into a final requirement. Source inspection alone does not establish that a requirement has passed testing.

2. Overall Description

2.1 Product Perspective

The application consists of a responsive web storefront, a restricted administration interface, a backend API and a relational database. It integrates with external services for online payments, image storage and postal PIN lookup.

2.2 User Classes

User class	Main responsibilities / capabilities
Guest	Browse and search products, view product information and create an account or sign in

User class	Main responsibilities / capabilities
Customer	Manage their own profile, cart, wishlist, addresses, orders and reviews
Administrator	Manage catalog information, inventory, order processing, user roles and operational information

Razorpay, Cloudinary and the postal lookup provider are external systems rather than human user roles.

2.3 Operating Environment

The current project uses React and Vite for the frontend, Node.js and Express for the backend, and PostgreSQL with Prisma for data storage. The customer interface is in English and prices are displayed in INR. Supported browser versions, deployment hardware and the final evaluation environment are TBD.

2.4 Assumptions and Dependencies

- Users need a network connection to retrieve current catalog data and perform account, cart and order operations.
- Online payment depends on Razorpay availability and valid configuration.
- Cloudinary image uploads depend on configured credentials; the project also supports local upload storage.
- Delivery PIN lookup depends on the postal provider. A resolved postal location does not establish courier serviceability.
- Administrative functions require an authenticated account with the administrator role.

2.5 Constraints and Operations

- The delivery region is India; the interface language is English and monetary values use INR.
- The initial release is a responsive browser application with customer and administrator interfaces.
- The project uses React, Node.js/Express, PostgreSQL and Prisma. These are existing implementation constraints rather than customer-facing requirements.

- No dedicated scanner, printer or other specialist hardware interface is required. Invoice printing uses browser facilities.
- Local development supports separate frontend/backend processes and a root development startup command. Deployment also includes Docker configuration.
- Hardware capacity, supported browser versions, backup schedule and recovery responsibilities require agreement before final approval.

3. Functional Requirements

3.1 Writing Convention

Give each requirement a unique ID and write it as an observable statement beginning with **"The system shall..."**. Keep one main behaviour per requirement. For each requirement, add its actor, preconditions, inputs, expected result, failure behaviour and acceptance criteria. These starter statements must be expanded and reviewed before submission.

3.2 Functional Requirement Register

All entries are **Draft**, with proposed **Must Have** priority for the current release. Acceptance checks describe expected results, not completed verification. Requirement IDs are retained across the register and worked examples.

ID	Module	Draft requirement
FR-01	Registration	The system shall allow a guest to register an account using valid registration details and an email address not already registered.
FR-02	Authentication	The system shall allow a registered user to sign in using valid credentials.
FR-03	Catalog	The system shall return products matching the customer's search and supported filter selections.
FR-04	Product details	The system shall display the selected product's stored description, images, price and availability.
FR-05	Cart	The system shall allow a customer to update a cart item's quantity within available stock limits.

ID	Module	Draft requirement
FR-06	Selected checkout	The system shall create an order containing only the cart items selected by the customer for checkout.
FR-07	Addresses	The system shall allow a customer to save a delivery address associated with their own account.
FR-08	Online payment	The system shall verify an online payment with the configured payment gateway before marking the order as paid.
FR-09	Order history	The system shall allow a customer to view orders belonging to their own account.
FR-10	Wishlist	The system shall allow a customer to save a product to their wishlist.
FR-11	Reviews	The system shall allow an authenticated customer to submit at most one review per product, with an integer rating from 1 to 5.
FR-12	Administration	The system shall restrict administrative operations to authenticated administrators.
FR-13	Delivery location	The system shall apply a delivery PIN location only after a successful postal lookup.

The detailed register below expands operations absent from the initial starter list. FR-06 is detailed in Section 3.4.

ID	Actor / operation	Requirement	Acceptance criterion / failure behaviour
FR-14	Account holder / logout	The system shall terminate the current authenticated session when the account holder signs out.	Protected operations require sign-in after logout; account-specific cart and cached private data are cleared from the interface.
FR-15	Customer / profile	The system shall allow an authenticated customer to update their own supported profile fields.	Valid name/phone changes persist; invalid fields produce a validation message. Another customer's record cannot be changed.

ID	Actor / operation	Requirement	Acceptance criterion / failure behaviour
FR-16	Account holder / password	The system shall allow an account holder to change their password after verifying their current password.	A new password requires at least eight characters, an uppercase letter and a number; an incorrect current password is rejected. Existing sessions are revoked on success.
FR-17	Guest or customer / catalog order	The system shall paginate catalog results using supported sorting choices.	Name, price and creation-date ordering follow the selected direction; page changes retain active filters.
FR-18	Customer / add cart item	The system shall add a requested product quantity to the customer's cart when sufficient stock exists.	The cart count and totals update; a quantity exceeding stock is rejected without a partial cart write.
FR-19	Customer / remove cart item	The system shall remove the selected product from the customer's cart.	The product no longer appears and totals reflect the remaining items.
FR-20	Customer / address update	The system shall allow a customer to edit an address belonging to their own address book.	Valid changes persist; attempts to update another account's address are rejected.
FR-21	Customer / address deletion	The system shall allow a customer to delete an address from their own address book.	The address is unavailable for future checkout while earlier order address snapshots remain intact.
FR-22	Customer / COD checkout	The system shall accept Cash on Delivery as a supported checkout method.	A successful COD order is confirmed with payment pending; Razorpay checkout is not required.
FR-23	Customer / payment retry	The system shall permit retrying payment for an	An incomplete or failed payment does not appear as

ID	Actor / operation	Requirement	Acceptance criterion / failure behaviour
		eligible unpaid online order.	paid; a retry uses the eligible existing application order.
FR-24	Customer / cancellation	The system shall allow a customer to cancel their own PENDING or CONFIRMED order.	Other fulfilment states reject customer cancellation; successful cancellation restores reserved stock once.
FR-25	Administrator / refund initiation	The system shall allow an administrator to initiate a refund for an eligible cancelled paid online order.	The attempt is recorded and remains pending until gateway confirmation; failed or uncertain submission is not shown as a completed refund.
FR-26	Payment provider / refund confirmation	The system shall reconcile refund status from authenticated gateway notifications.	A confirmed completion updates refund/payment records; duplicate or delayed notifications do not reverse completed state. Invalid signatures are rejected.
FR-27	Administrator / COD collection	The system shall allow an administrator to record cash collection for an eligible delivered COD order.	Collection marks payment paid and records the collection time and administrator; repeated collection does not record a second payment.
FR-28	Customer / wishlist removal	The system shall remove a saved product from the customer's wishlist on request.	The product disappears from the wishlist and saved indicators refresh.
FR-29	Customer / review retrieval	The system shall display product reviews with pagination and aggregate ratings.	Review pages belong to the selected product; rating counts cover all its reviews rather than only the visible page.
FR-30	Administrator / catalog maintenance	The system shall allow an administrator to	Valid names, descriptions, category assignments, prices

ID	Actor / operation	Requirement	Acceptance criterion / failure behaviour
		create and update product records.	and stock persist; invalid inputs return field errors.
FR-31	Administrator / product deletion	The system shall allow product deletion only when historical data constraints permit it.	A product referenced by order history cannot be silently removed from that history; a blocked deletion gives a useful error.
FR-32	Administrator / image management	The system shall allow an administrator to upload supported product images.	Accepted files become product image references; unsupported or excessive uploads return validation errors.
FR-33	Administrator / categories	The system shall allow an administrator to maintain catalog categories.	Category names and slugs follow uniqueness rules; invalid references do not leave orphan products.
FR-34	Administrator / fulfilment	The system shall restrict order status changes to permitted transitions.	Invalid transitions are rejected; unpaid online orders cannot advance through paid-order fulfilment.
FR-35	Administrator / user directory	The system shall provide a paginated user directory to administrators.	Customer sessions cannot retrieve the directory; supported role changes affect subsequent authorization.
FR-36	Administrator / operational overview	The system shall display order, revenue, user, product and low-stock information.	Values come from stored records; failures appear as actionable errors rather than invented zero values.
FR-37	Customer / order search	The system shall allow customers to search and filter their own order history.	Product-name or order-number searches, date presets and supported status filters retain account isolation.
FR-38	Customer / order details	The system shall display a customer's order	The two status types are distinct; another customer's

ID	Actor / operation	Requirement	Acceptance criterion / failure behaviour
		items, total, address snapshot, fulfilment state and payment state.	order identifier cannot expose private details.
FR-39	Customer / buy again	The system shall allow a customer to add an available previously ordered product to their current cart.	Current stock and price apply; the historic order is unchanged.

3.3 Acceptance Criteria for Initial Requirements

Requirement	Inputs / trigger	Expected result and failure behaviour
FR-01	Name, email, password, optional phone	A valid unique account is created; duplicate email and invalid fields are rejected. Password rules are defined by FR-16.
FR-02	Registered email and password	Valid credentials establish an authenticated session; invalid credentials do not expose private data.
FR-03	Search text, category, price and stock filters	Returned products satisfy the requested criteria; no matches produce an explicit empty state.
FR-04	Product slug	Stored product information appears; an unknown slug produces a not-found state and unavailable images use a fallback.
FR-05	Product and positive integer quantity	The quantity and subtotal update within stock limits; invalid/excess quantities are rejected.
FR-07	Recipient, phone, address, city, state, PIN and country	A valid address is stored for the signed-in account; missing/invalid required values are rejected.

Requirement	Inputs / trigger	Expected result and failure behaviour
FR-08	Application order and Razorpay verification values	Only valid gateway/order/amount verification can mark payment paid; a duplicate confirmation does not duplicate stock or payment effects.
FR-09	Signed-in customer and history request	Paginated results contain only that customer's orders; guests require authentication.
FR-10	Product identifier	The item is saved once per account; a repeated toggle updates the saved state without duplicates.
FR-11	Product, rating and optional comment	A rating must be an integer from 1 to 5; a second review by the same account for that product is rejected.
FR-12	Administrative request	An authenticated administrator is authorized; guests and customer roles are rejected.
FR-13	Six-digit Indian PIN	Apply is enabled after a successful lookup; invalid codes and provider failures give an error. Postal validity does not imply shipping serviceability.

3.4 Worked Example: FR-06 — Selected Cart Checkout

Field	Specification
Actor	Customer
Preconditions	The customer is signed in, has cart items and has a saved delivery address.
Inputs	Selected cart item IDs, delivery address and payment method
Trigger	The customer submits the checkout order.
Expected result	An order is created from the selected items using server-side selling prices.

Field	Specification
Cart effect	Purchased rows are removed; unchecked rows remain in the cart.
Stock effect	Stock is reduced only for ordered quantities.
Failure behaviour	An empty selection, invalid selection, invalid address or insufficient selected-item stock is rejected without completing an order.

Proposed acceptance criteria:

1. Given cart products A and B, when only A is selected and ordered, the new order contains A and excludes B.
2. B remains in the cart and its stock is unchanged.
3. The order total equals the sum of the selected quantities multiplied by their server-side selling prices.
4. With no items selected, the interface disables checkout and an explicit empty selection sent to the API is rejected.
5. If selected stock becomes insufficient, no order or partial inventory reduction is committed.

The API also supports legacy whole-cart checkout when item IDs are omitted. Distinguish this from an explicit empty selection when documenting interface behaviour.

4. External Interface Requirements

Interface	What to document
Customer interface	Screens, navigation, form fields, validation, loading and error states, and responsive behaviour
Administration interface	Restricted screens and permitted actions for each administrative function
Backend API	Required input, authentication, output and error behaviour for important operations
Razorpay	Payment creation, verification, cancellation/failure handling and refund status confirmation
Cloudinary / local uploads	Supported image inputs, upload results and fallback behaviour

Interface	What to document
Postal PIN provider	Six-digit PIN input, resolved location, invalid PINs and provider failure behaviour

4.1 Interface Contracts

ID	Interface	Input / communication	Required response or failure handling
INT-01	Browser storefront	Keyboard, pointer or touch interactions on public and account screens	Search, cart, checkout and account forms shall expose loading, empty and failure states. Private actions require sign-in.
INT-02	Administration	Authenticated administrator navigation and validated form submissions	Unauthorized roles shall be rejected at the API as well as restricted in the interface.
INT-03	Application API	JSON over HTTP in local development; HTTPS required for production access	Input validation shall report invalid requests; private operations shall require authentication and enforce ownership.
INT-04	Image upload	Multipart file upload by an administrator	Supported files shall return stored image references; local upload fallback depends on deployment configuration.
INT-05	Razorpay checkout	Gateway order creation, customer checkout callback and server verification	Failed, dismissed and incomplete payment attempts shall remain distinguishable from verified payment.
INT-06	Razorpay webhook	Signed gateway event delivered to the configured webhook endpoint	Valid events shall reconcile payment/refund state idempotently; transient processing failures shall allow provider retry.

ID	Interface	Input / communication	Required response or failure handling
INT-07	Postal lookup	Six-digit PIN through the backend location endpoint	Successful resolution shall provide city/district and state; timeout or invalid PIN shall not commit an unverified location.

Detailed field schemas are maintained in `Backend/src/validators`. API examples and UI screenshots may be added as submission appendices after the requirement wording is reviewed.

5. Non-Functional Requirements

These are draft requirements to review and verify. Do not describe them as achieved merely because they appear in this document.

ID	Quality	Draft requirement / target	Verification approach
NFR-01	Security	The system shall reject unauthorised access to another customer's private account resources.	Attempt cross-account reads and writes for addresses, cart and orders.
NFR-02	Security	The system shall reject administrative API requests made with a customer account.	Exercise administrative endpoints with a customer session.
NFR-03	Data integrity	The system shall prevent inventory from falling below zero when competing orders request the same stock.	Submit concurrent orders against a controlled stock fixture.
NFR-04	Error handling	The system shall display an actionable error when a customer operation fails.	Simulate validation, network and API failures for each main flow.
NFR-05	Responsive usability	The system shall keep primary shopping controls usable at agreed phone and desktop viewport sizes.	Specify viewport sizes and inspect main flows; sizes are TBD.

ID	Quality	Draft requirement / target	Verification approach
NFR-06	Performance	Response-time percentile, maximum response time, concurrent load and test environment are TBD.	Define a measurable target, then run a repeatable load test.

5.1 Additional Quality Requirements

These requirements and evaluation conditions remain proposed until reviewed by the project guide. Numerical targets below are evaluation proposals, not measured application results.

ID	Quality	Draft requirement / target	Verification approach
NFR-07	Keyboard usability	The system shall expose primary search, cart, address, checkout and sign-in actions through keyboard-operable controls with visible focus.	Complete each main flow using Tab, Shift+Tab, Enter and Space; check focus returns after dialogs close. This is not a claim of full accessibility-standard compliance.
NFR-08	Credential protection	The system shall exclude password values and reusable authentication tokens from ordinary application logs and public API responses.	Inspect representative responses/logs and stored credential representations using controlled test accounts.
NFR-09	Transaction integrity	The system shall commit checkout order creation, selected-item removal and inventory reservation as one transaction.	Force a failure during checkout and confirm that all associated database changes roll back.
NFR-10	Traceability of failures	The system shall attach a request identifier to backend request processing so a reported	Trigger a controlled API failure and locate the corresponding log by request identifier.

ID	Quality	Draft requirement / target	Verification approach
		API failure can be correlated with its server log.	
NFR-11	Compatibility	The system shall support the agreed browser matrix for the primary shopping and administrative workflows.	Proposed evaluation: Chrome, Edge and Firefox desktop; Chrome on Android. Record exact versions before evaluation. Support remains unapproved until confirmed.
NFR-12	Recovery	The deployed system shall have a documented database backup and recovery procedure.	Agree backup owner, frequency, retention, maximum acceptable data loss and restoration time; perform and record a controlled restore before claiming readiness.

For NFR-05, proposed evaluation widths are 390 CSS pixels for phones and 1440 CSS pixels for desktop, with readable content and no page-wide horizontal overflow. Product carousels may scroll within their own containers. These widths require review and do not replace evaluation on actual devices.

For NFR-06, the guide must approve a response-time percentile, load profile, dataset size, hardware/network configuration and treatment of third-party latency. Availability targets, maintenance windows and recovery thresholds are also open decisions; no uptime percentage or load-test result is asserted here.

6. Data Requirements

The main business data includes users, categories, products, product images, carts, cart items, addresses, orders, order items, payments, refund attempts, wishlists and reviews. Authentication also uses refresh-token records.

6.1 Initial Data Dictionary

Entity	Important data	Required relationship / integrity rule
User	Name, unique email, password hash, role, session version	Account ownership anchors private records; credentials shall not appear in public responses.
Category	Name, unique slug	Products reference an existing category.
Product	Name, slug, description, selling price, optional original price, discount, stock	Money uses decimal storage; inventory allocation shall not oversell stock.
ProductImage	URL and optional source/provenance metadata	Each image references a product.
Cart / CartItem	Account, product, quantity	One cart per user and one row per product within a cart.
Address	Recipient, phone, address lines, city, state, postal code, country	Each saved address belongs to one user.
Order / OrderItem	User, total, fulfilment state, payment state, delivery snapshot, ordered quantities/prices	Order-item prices and the shipping snapshot preserve purchase-time data.
Payment	Application order, amount, provider, payment status, gateway identifiers	One payment record per application order; gateway payment/order identifiers are unique when present.
RefundAttempt	Payment, attempt number, request key, gateway refund ID, amount, status	Attempts are recorded individually and do not become completed solely because submission was requested.
Wishlist	User and product	Each user/product pair is unique.
Review	User, product, rating, comment	Each user/product pair is unique; rating validation permits

Entity	Important data	Required relationship / integrity rule
		1–5.
RefreshToken	Token hash, user, expiry and revocation metadata	Raw refresh tokens shall not be stored as database token values.

6.2 Field Types and Business Rules

Data / rule	Type or constraint	Required behaviour
Identifiers and quantities	Integer identifiers; positive integer purchase quantities	Unrecognized/private identifiers shall not grant access; invalid quantities shall be rejected.
Product and transaction amounts	Decimal values with two stored fractional digits	Order totals shall be calculated from server-side selling prices and quantities. Browser-submitted totals are not authoritative.
Product name and description	Required strings	Catalog forms shall validate required values before a write is accepted.
Original price	Optional decimal	Crossed-out MRP and a discount label shall appear only when a valid original price exceeds the selling price.
Inventory	Integer stock	Allocation shall check available stock atomically; successful cancellation shall restore an allocation no more than once.
Customer address	Required recipient/contact/location fields as validated by the API	An order shall retain the address snapshot captured at checkout, even if the address book later changes.
Fulfilment and payment	Separate enumerated states	An unpaid online order shall not be described as paid; fulfilment

Data / rule	Type or constraint	Required behaviour
states		cancellation and refund completion shall remain distinct.
Reviews	Integer rating 1–5; optional comment	The user/product pair shall remain unique.
Gateway identifiers	Optional strings until supplied by the gateway	Once assigned, identifiers shall refer to the same application payment rather than being reused for unrelated orders.
Refund attempts	Payment reference, positive attempt number, request key and status	A submission attempt shall remain distinguishable from gateway-confirmed success.

6.3 Ownership and Relationships

Parent	Related records	Cardinality / ownership
User	Cart	Zero or one cart per account; created when needed.
User	Addresses, orders, reviews, wishlist rows, refresh tokens	One user to many related records; private resources enforce account ownership.
Category	Products	One category to many products; each product has one category.
Product	Images, cart rows, order rows, reviews, wishlist rows	One product to many related records. Historical order references constrain deletion.
Cart	CartItem	One cart to many items; each product occurs once within that cart.
Order	OrderItem	One order to many purchased lines with quantity and price snapshots.
Order	Payment	Zero or one payment record per order.
Payment	RefundAttempt	One payment to many recorded submission attempts.

This relationship table is the draft basis for an ER diagram. The diagram and detailed field lengths should be finalized against `Backend/prisma/schema.prisma` when preparing submission figures. Data-retention periods, account-erasure procedures and backup retention require a policy decision; no automatic deletion schedule is assumed.

7. Use Cases and System Models

The following use cases describe proposed observable behaviour. Preconditions and postconditions specify successful operation; alternative flows explain expected failures.

ID	Use case	Primary actor
UC-01	Purchase selected cart items	Customer
UC-02	Search and filter products	Guest / customer
UC-03	Maintain a delivery address	Customer
UC-04	Cancel an eligible order	Customer
UC-05	Maintain product information	Administrator
UC-06	Advance order fulfilment status	Administrator
UC-07	Register, sign in and maintain an account	Guest / account holder
UC-08	Manage a shopping cart	Customer
UC-09	Maintain a wishlist	Customer
UC-10	Read and submit product reviews	Guest / customer
UC-11	View order history and buy again	Customer

ID	Use case	Primary actor
UC-12	Administer payments, users and operations	Administrator / payment provider

7.1 UC-01 — Purchase Selected Cart Items

Goal: Place an order for the chosen cart items. **Preconditions:** The customer is signed in, has cart items and owns the selected delivery address. **Trigger:** The customer proceeds to checkout.

1. The customer selects cart items and chooses or creates a delivery address.
2. The customer chooses online payment or COD and reviews the selected items and total.
3. The server validates ownership, selections, current prices and available stock.
4. The server creates the order, reserves stock and removes only purchased cart items in one transaction.
5. For COD, the order is confirmed with payment pending.
6. For online payment, the application prepares a gateway order and opens Razorpay checkout. The order remains unpaid until verified payment is recorded.
7. The confirmation view reports the actual order and payment states.

Alternatives: An empty selection, unauthorized address or insufficient stock rejects checkout without partial database changes. Dismissed/failed online payment retains an unpaid application order for an eligible retry. A repeated checkout request with the same supported request key and identical inputs shall return the same order; different inputs with that key are rejected.

Postconditions: A successful application order stores item-price and address snapshots; unchecked items remain in the cart. Payment completion is recorded separately from order creation. **Requirements:** FR-06–08, FR-22–23; NFR-03 and NFR-09.

7.2 UC-02 — Search and Filter Products

Goal: Find products matching the shopper's criteria. **Preconditions:** The catalog is reachable. **Trigger:** A guest/customer enters search text, chooses filters or changes sorting.

1. The shopper submits search text or selects a supported category, price or stock filter.
2. The system requests the matching page using the selected sorting order.
3. The shopper navigates result pages and opens a product detail page.

Alternatives: No matches produces an empty state. An unknown product produces a not-found state. A network/rate-limit failure displays an error and a supported retry/recovery path. Missing images use a fallback.

Postconditions: The visible results and product information correspond to the selected request. **Requirements:** FR-03–04, FR-17; NFR-04–05.

7.3 UC-03 — Maintain a Delivery Address

Goal: Maintain addresses available for future checkout. **Preconditions:** The customer is signed in. **Trigger:** The customer opens the address book.

1. The system lists addresses owned by the current account.
2. The customer creates, edits or deletes an address.
3. The system validates the required fields and ownership, saves a valid change, and refreshes the address list.

Alternatives: Invalid fields are identified without accepting the write. A resource belonging to another account is rejected. A failed request leaves the customer able to correct or retry it.

Postconditions: Future checkout uses the updated address book; existing order snapshots retain purchase-time information. The navbar's verified PIN selection remains a separate display preference. **Requirements:** FR-07, FR-13, FR-20–21; NFR-01.

7.4 UC-04 — Cancel an Eligible Order

Goal: Cancel an order before the customer cancellation cutoff. **Preconditions:** The customer owns an order in PENDING or CONFIRMED state. **Trigger:** The customer confirms the cancellation action.

1. The system validates ownership and current order state.
2. The system cancels the order and restores reserved quantities once.

3. If an online payment has already been captured, the system records that a refund is pending.
4. The interface displays the current cancellation and payment/refund states.

Alternatives: Customer cancellation of PROCESSING, SHIPPED or DELIVERED orders is rejected. Repeating an already completed cancellation does not restore stock twice. A concurrent payment or administrator update is reconciled against the current order state.

Postconditions: A cancelled order cannot resume normal fulfilment. Refund initiation and gateway confirmation follow UC-12; cancellation alone does not establish completed repayment. **Requirements:** FR-24–26; NFR-03.

7.5 UC-05 — Maintain Product Information

Goal: Maintain accurate catalog entries. **Preconditions:** The actor is an authenticated administrator. **Trigger:** The administrator opens catalog maintenance.

1. The administrator selects a product or starts a new entry.
2. The administrator supplies product details, price, stock and category; supported images may be uploaded.
3. The server validates the request and persists the permitted changes.
4. The relevant catalog data refreshes after a successful write.

Alternatives: Invalid pricing/stock fields, unsupported images or invalid category references are rejected. Deleting a product referenced by order history is blocked with a useful error.

Postconditions: Authorized catalog changes are available to subsequent catalog requests; historical order price snapshots remain intact. **Requirements:** FR-12, FR-30–33.

7.6 UC-06 — Advance Order Fulfilment

Goal: Record an order's operational progress. **Preconditions:** The actor is an administrator and the requested order exists. **Trigger:** The administrator submits a status update.

1. The system displays the current fulfilment and payment states.
2. The administrator chooses a permitted next state.

3. The server verifies the transition and any required payment condition before saving it.
4. The order view reflects the resulting state.

Alternatives: Invalid transitions, unauthorized requests and attempts to advance unpaid online orders are rejected. Administrator cancellation follows the permitted state model and restores stock only once.

Postconditions: A valid transition is recorded without skipping the applicable guards. **Requirements:** FR-12, FR-34.

7.7 Supporting Use Cases

Use case	Preconditions / trigger	Main flow	Alternative and postcondition
UC-07	Guest opens registration/sign-in, or account holder opens settings	Submit validated registration/credentials; manage supported profile fields, password and sign-out.	Duplicate email, invalid credentials or incorrect current password are rejected. Password change revokes existing sessions; sign-out clears the current session.
UC-08	Customer chooses cart actions	Add products, change quantity and remove items; totals and selection state update.	Insufficient stock or invalid quantities reject the write; an empty cart cannot produce a selected checkout.
UC-09	Customer selects a wishlist control	Save or remove the chosen product; update saved indicators.	Unauthorized private actions require sign-in; repeated toggles do not create duplicate rows.

Use case	Preconditions / trigger	Main flow	Alternative and postcondition
UC-10	Shopper opens product reviews	Read review pages and aggregate ratings; a signed-in customer may submit a 1–5 rating with optional comment.	Duplicate user/product reviews and invalid ratings are rejected; reading another review page does not change full-product aggregate counts.
UC-11	Customer opens order history	Search/filter owned orders, inspect details and add an available previous item to the current cart.	A private order belonging to someone else is not disclosed. Reordering uses current stock and selling price.
UC-12	Administrator opens payment/user/operations controls; provider sends a signed event	Submit eligible refunds, reconcile gateway results, record delivered COD collection, review users/roles and operational summaries.	Invalid signatures, customer-role requests and ineligible payment actions are rejected. Refund submission is not reported as completed repayment; repeated confirmed events do not duplicate effects.

7.8 Fulfilment State Model

Current state	Permitted next states	Conditions
PENDING	CONFIRMED, CANCELLED	Online confirmation requires verified paid status. Customer cancellation is permitted.
CONFIRMED	PROCESSING, CANCELLED	Customer cancellation is permitted before progression.
PROCESSING	SHIPPED, CANCELLED	Cancellation at this stage is an administrator operation.
SHIPPED	DELIVERED	Customer cancellation is unavailable.
DELIVERED	None	Delivered COD orders can have their payment collection recorded separately.
CANCELLED	None	Refund/payment reconciliation may continue; fulfilment does not reopen.

This table describes fulfilment only. Payment and refund states are separate. A use-case figure, ER diagram and checkout activity/DFD can be added when the department confirms the required diagram types; the relationship and transition tables above supply their source content.

8. Acceptance Criteria and Traceability

Traceability links each requirement to a use case and an observable acceptance check. All rows below have status **Not assessed for this SRS**. Historical automated test results do not automatically approve these document requirements; record dated evidence separately after evaluation.

Requirement	Use case	Proposed acceptance check
FR-01	UC-07	AT-F01: Register valid details; reject duplicate email and invalid fields.
FR-02	UC-07	AT-F02: Accept correct credentials; reject incorrect credentials.
FR-03	UC-02	AT-F03: Combine supported search/filter inputs and inspect matching results.

Requirement	Use case	Proposed acceptance check
FR-04	UC-02	AT-F04: Open an existing and unknown product; inspect stored details and fallback states.
FR-05	UC-08	AT-F05: Update a valid cart quantity and reject an excessive quantity.
FR-06	UC-01	AT-F06: Order selected items while preserving unchecked items and their inventory.
FR-07	UC-03	AT-F07: Save a valid address under the current account only.
FR-08	UC-01	AT-F08: Verify a valid gateway payment; reject altered verification values.
FR-09	UC-11	AT-F09: List only the signed-in customer's orders.
FR-10	UC-09	AT-F10: Save a product and verify no duplicate wishlist row.
FR-11	UC-10	AT-F11: Submit a valid review; reject invalid ratings and a second review.
FR-12	UC-05, UC-06, UC-12	AT-F12: Compare administrator, customer and guest access to protected operations.
FR-13	UC-03	AT-F13: Resolve a valid PIN and reject invalid/failed lookup application.
FR-14	UC-07	AT-F14: Sign out and confirm private actions require authentication.
FR-15	UC-07	AT-F15: Save valid profile changes and reject another account's update.
FR-16	UC-07	AT-F16: Reject an incorrect current password; verify old sessions are revoked after a valid change.
FR-17	UC-02	AT-F17: Change supported sort orders and pages while retaining filters.
FR-18	UC-08	AT-F18: Add an available quantity; reject a quantity exceeding stock.

Requirement	Use case	Proposed acceptance check
FR-19	UC-08	AT-F19: Remove an item and verify remaining totals.
FR-20	UC-03	AT-F20: Edit an owned address; reject unauthorized editing.
FR-21	UC-03	AT-F21: Delete a saved address and verify an earlier order snapshot is retained.
FR-22	UC-01	AT-F22: Place a COD order and verify confirmed fulfilment with payment pending.
FR-23	UC-01	AT-F23: Retry an eligible unpaid order without creating a second application order.
FR-24	UC-04	AT-F24: Cancel an eligible owned order, repeat the request and verify a single stock restoration.
FR-25	UC-12	AT-F25: Submit an eligible refund and verify a recorded pending attempt.
FR-26	UC-12	AT-F26: Reconcile valid repeated/refund events and reject an invalid signature.
FR-27	UC-12	AT-F27: Record collection for delivered COD; reject ineligible collection and duplicate payment effects.
FR-28	UC-09	AT-F28: Remove a saved product and verify saved indicators refresh.
FR-29	UC-10	AT-F29: Change review pages while preserving full-product rating counts.
FR-30	UC-05	AT-F30: Create/update valid product fields and reject invalid input.
FR-31	UC-05	AT-F31: Attempt to delete a historically ordered product without losing order references.
FR-32	UC-05	AT-F32: Upload a supported image and reject invalid upload input.
FR-33	UC-05	AT-F33: Maintain categories while preserving uniqueness and product references.

Requirement	Use case	Proposed acceptance check
FR-34	UC-06	AT-F34: Accept a permitted status change and reject an invalid/unpaid advancement.
FR-35	UC-12	AT-F35: Browse user pages as administrator and verify role-change authorization takes effect.
FR-36	UC-12	AT-F36: Compare summaries to controlled stored records and expose API failures.
FR-37	UC-11	AT-F37: Search/filter owned order pages without exposing another customer's orders.
FR-38	UC-11	AT-F38: Inspect item/address snapshots and separate payment/fulfilment states.
FR-39	UC-11	AT-F39: Buy again using current stock/price while preserving the earlier order.
NFR-01	UC-03, UC-08, UC-11	AT-N01: Attempt cross-account private-resource reads and writes.
NFR-02	UC-05, UC-06, UC-12	AT-N02: Exercise administrative APIs with a customer account.
NFR-03	UC-01, UC-04	AT-N03: Submit competing purchases against controlled stock.
NFR-04	UC-01–12	AT-N04: Trigger representative validation/network/API failures and inspect recovery guidance.
NFR-05	UC-01–12	AT-N05: Inspect main workflows at approved phone and desktop widths.
NFR-06	UC-02, UC-01	AT-N06: Execute the approved load profile; record latency percentiles and environment. Target pending.
NFR-07	UC-01, UC-02, UC-03, UC-07, UC-08	AT-N07: Complete primary workflows by keyboard and inspect focus.
NFR-08	UC-07, UC-12	AT-N08: Inspect logs and responses for exposed credentials/tokens.

Requirement	Use case	Proposed acceptance check
NFR-09	UC-01	AT-N09: Force a checkout failure and confirm complete transaction rollback.
NFR-10	UC-01–12	AT-N10: Correlate a controlled failed request with its request ID in server logs.
NFR-11	UC-01–12	AT-N11: Evaluate the approved browser/version matrix. Matrix pending approval.
NFR-12	UC-12	AT-N12: Restore a controlled backup using the agreed recovery procedure. Recovery thresholds pending.

For executed checks, add an evidence record containing the acceptance-check ID, date, evaluator, application revision, environment, result and evidence location. Do not replace a pending target with a pass/fail claim until its acceptance condition is agreed.

9. Open Decisions and Completion Checklist

- ☐ Fill the college, student, guide and academic-year details.
- ☐ Confirm the final project title and submission/page-format instructions.
- ☐ Review the scope inclusions and proposed exclusions with the project guide.
- ☐ Expand every feature into uniquely numbered, testable requirements.
- ☐ Confirm cancellation eligibility and the complete refund workflow against current behaviour; requesting a refund is different from gateway-confirmed completion.
- ☐ Complete interfaces, the data dictionary, use cases and required diagrams.
- ☐ Agree measurable quality targets and the evaluation environment.
- ☐ Map each requirement to acceptance criteria and verification evidence.
- ☐ Resolve all TBD entries, review consistency and update the revision history.
- ☐ Prepare the final Word/PDF layout and table of contents when the content is ready.

9.1 Decision Register

ID	Decision required	Current position	Suggested reviewer
OD-01	Submission identity	Student, roll number, college, department, guide and academic year are placeholders.	Student / guide
OD-02	Final scope and title	E-Commerce Web Application is the working title; confirm exclusions and proposed priorities.	Student / guide
OD-03	College format	Standard outline is in use; chapter order, page limit and required diagram types need confirmation.	Guide
OD-04	Quality evaluation	Phone/desktop widths and browser matrix are proposed; performance and availability targets remain unagreed.	Guide / project team
OD-05	Data operations	Backup ownership, retention, restoration targets and account-erasure procedure require a policy.	Project team / deployment owner
OD-06	Acceptance evidence	Requirement checks are specified but not assessed for this document.	Evaluator / project team

9.2 Review and Approval

Role	Name	Review date	Decision / signature
Student / project team	[To be filled]	[To be filled]	Pending
Project guide	[To be filled]	[To be filled]	Pending
Department reviewer, if required	[To be filled]	[To be filled]	Pending

The document remains a working draft until the required reviewers approve its content and unresolved decisions are recorded.